Modelling System Requirements

Modeling Methods

How to structure the problem?

The answer is just Simple, use **MODELS**

Model

Is a presentation of reality. Just a picture is worth a thousand of words, most system models are pictorial representations of reality.

SAMPLE FLOOR PLAN

Software System Models

- System modeling is the process of developing abstract models of a system, which **each model representing a different view or perspective of that system.**
- A model can be used as a tool for communication with clients and also among the development team members.
- Models are used during requirement engineering process to help derive the requirements of the system, during the design process to describe system components and also to document a system.

Models

Logical Models

Show what a system is or does?

Physical Models

Show not only what a system is or does? But also how the system is implemented.

Models used in structured methods and OO Processes

- Structured Methods - eg. SSADM

  Eg. Models – DFD, ERD, FHD etc.

- Object oriented Processes - eg. RUP

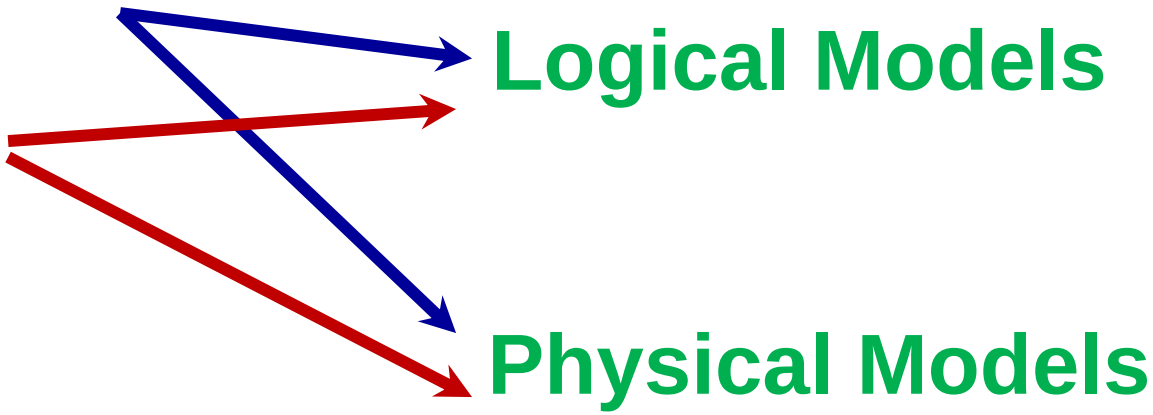
  Eg. UML Models – Class Diagrams,

  Sequence Diagrams etc.

Models

Models are divided into

- **Process Models**
- **Data Models**



Process Modeling

Process Modeling

Is a technique for organizing and documenting the Process Requirements and Design for a system.

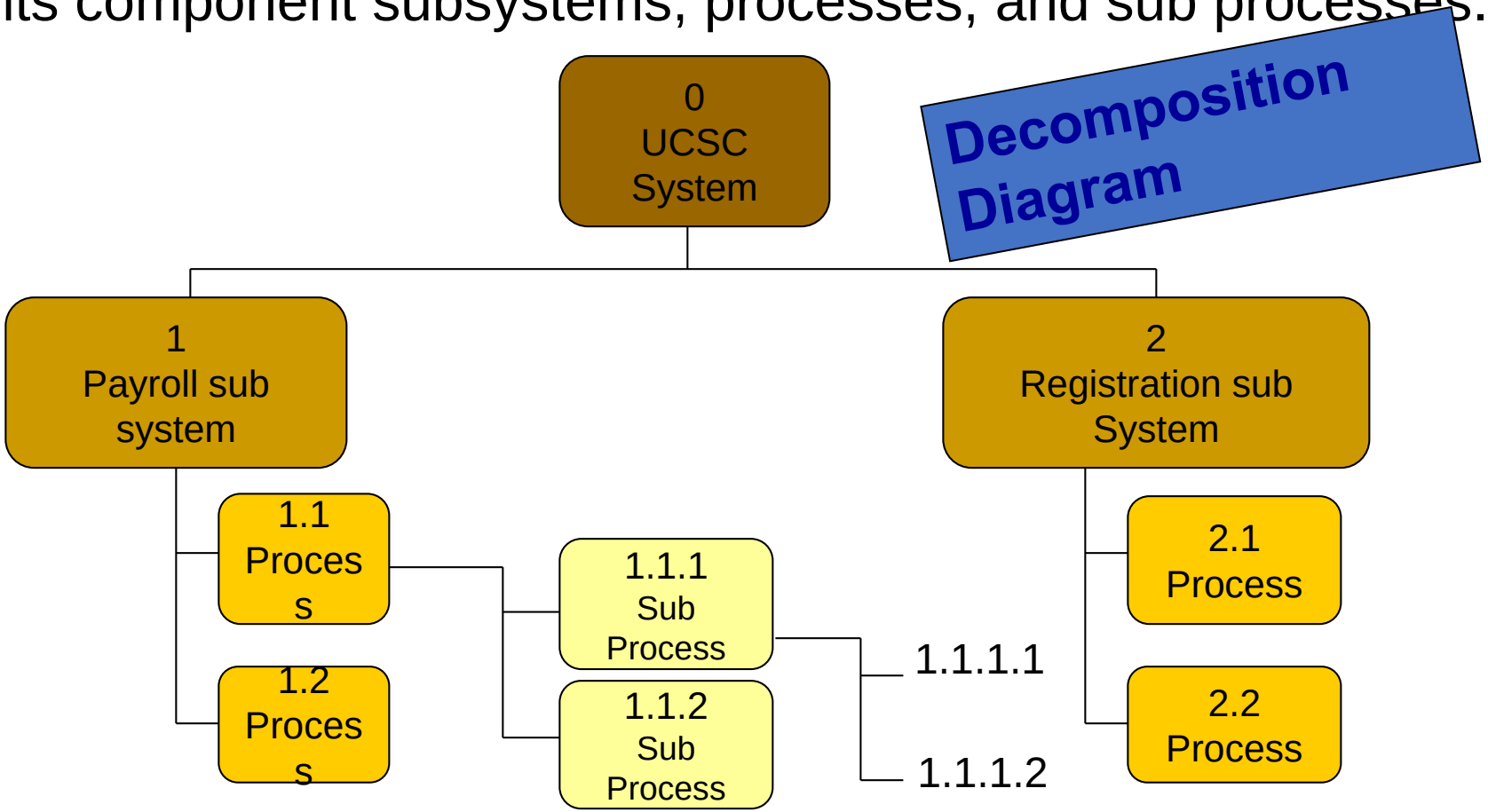
Data Flow Diagram : Popular Process Modeling Technique. (**Earlier, Not so popular now**)

Shows the flow of data through the system and the processing performed by the system.

Data Flow Diagram (DFD)

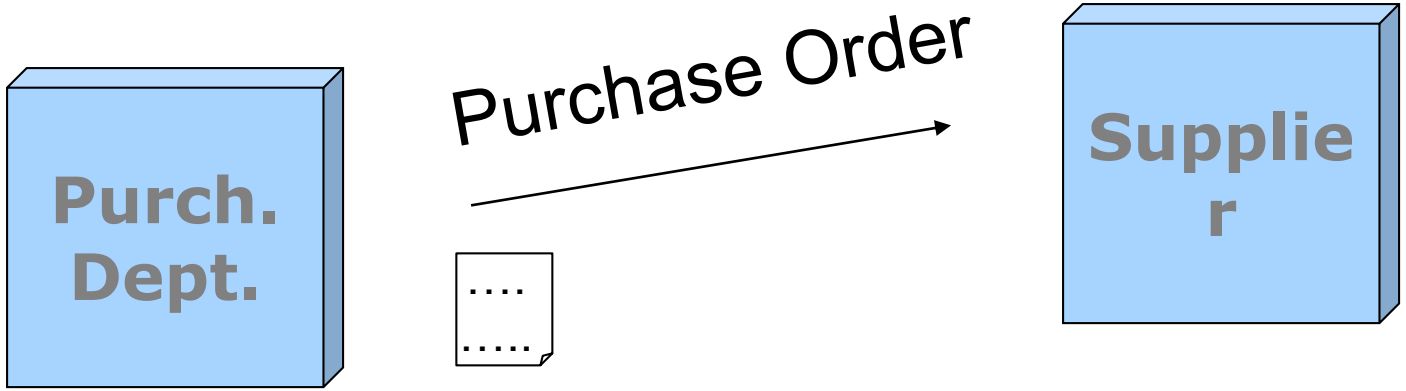
- Process Modeling Technique.
- Shows flow of data through a system and the processing performed by the system.
- Some analysts draw a **Decomposition diagram or Document flow diagram** before DFD.

Process Decomposition : Is a act of breaking a system into its component subsystems, processes, and sub processes.



Document Flow Diagrams

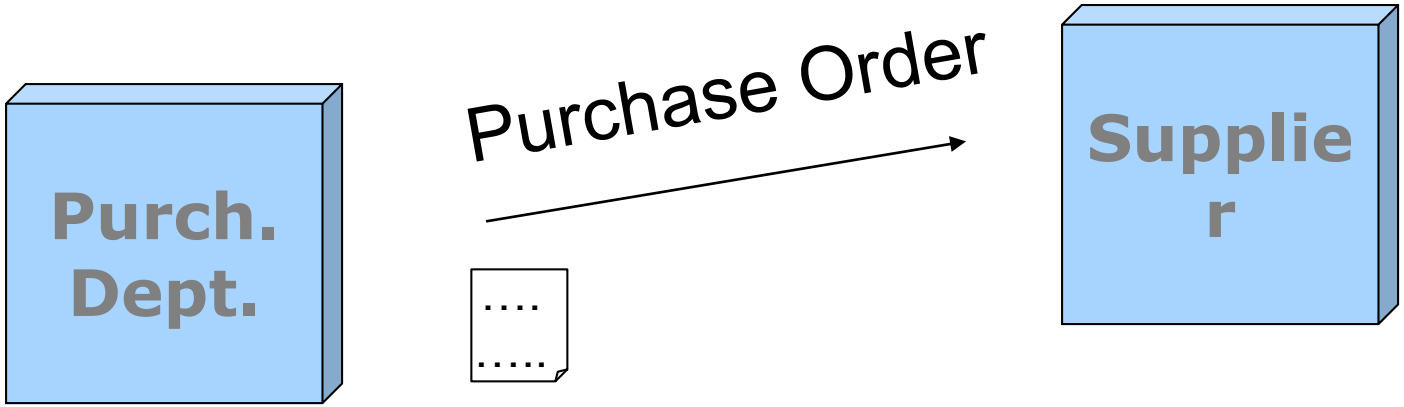
- Used to identify physical movement of documents.



Document Flow Diagrams

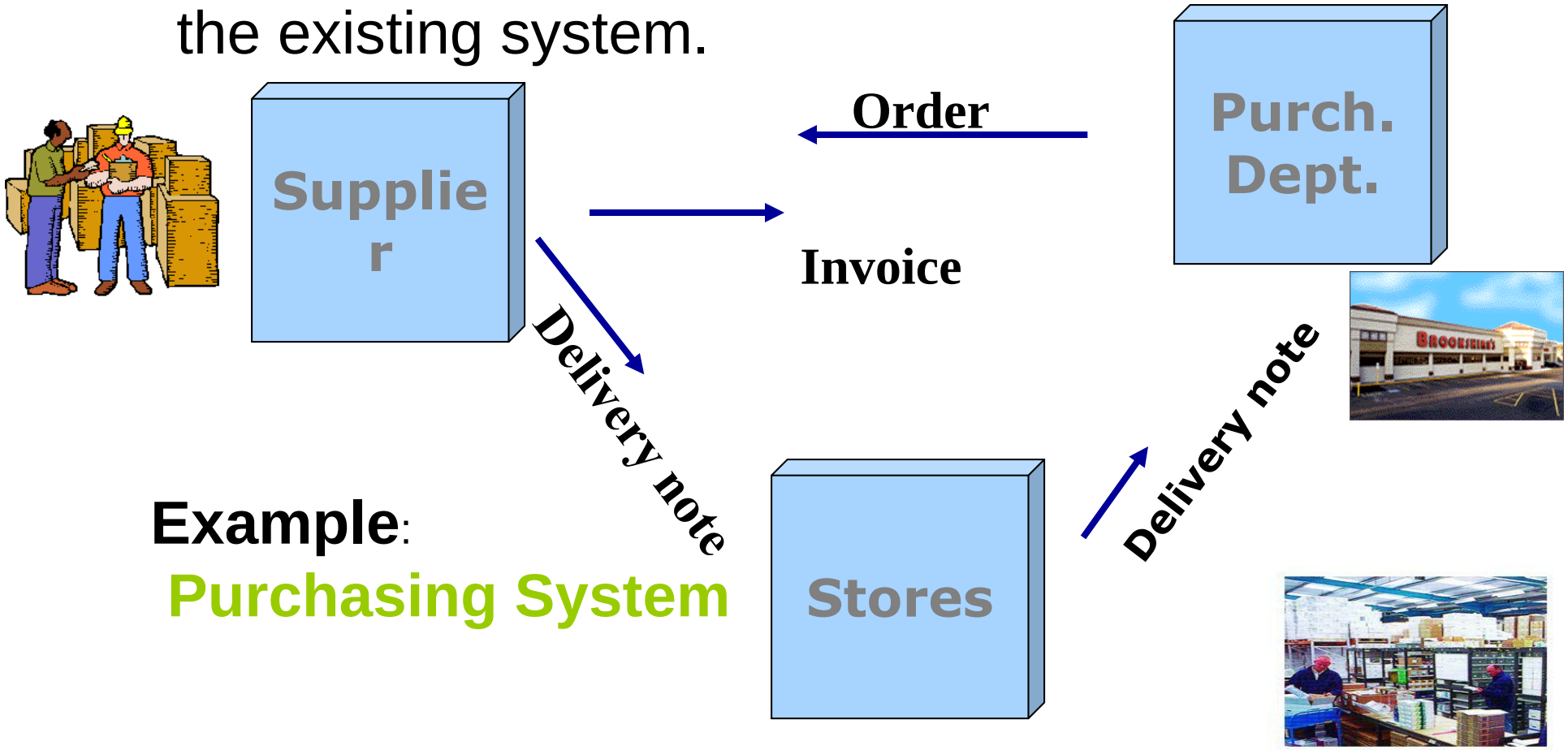
Shows

- where the document comes from,
- where it goes to , and
- what it is called.



Document Flow Diagrams

Used to examine the flow of documents within the existing system.

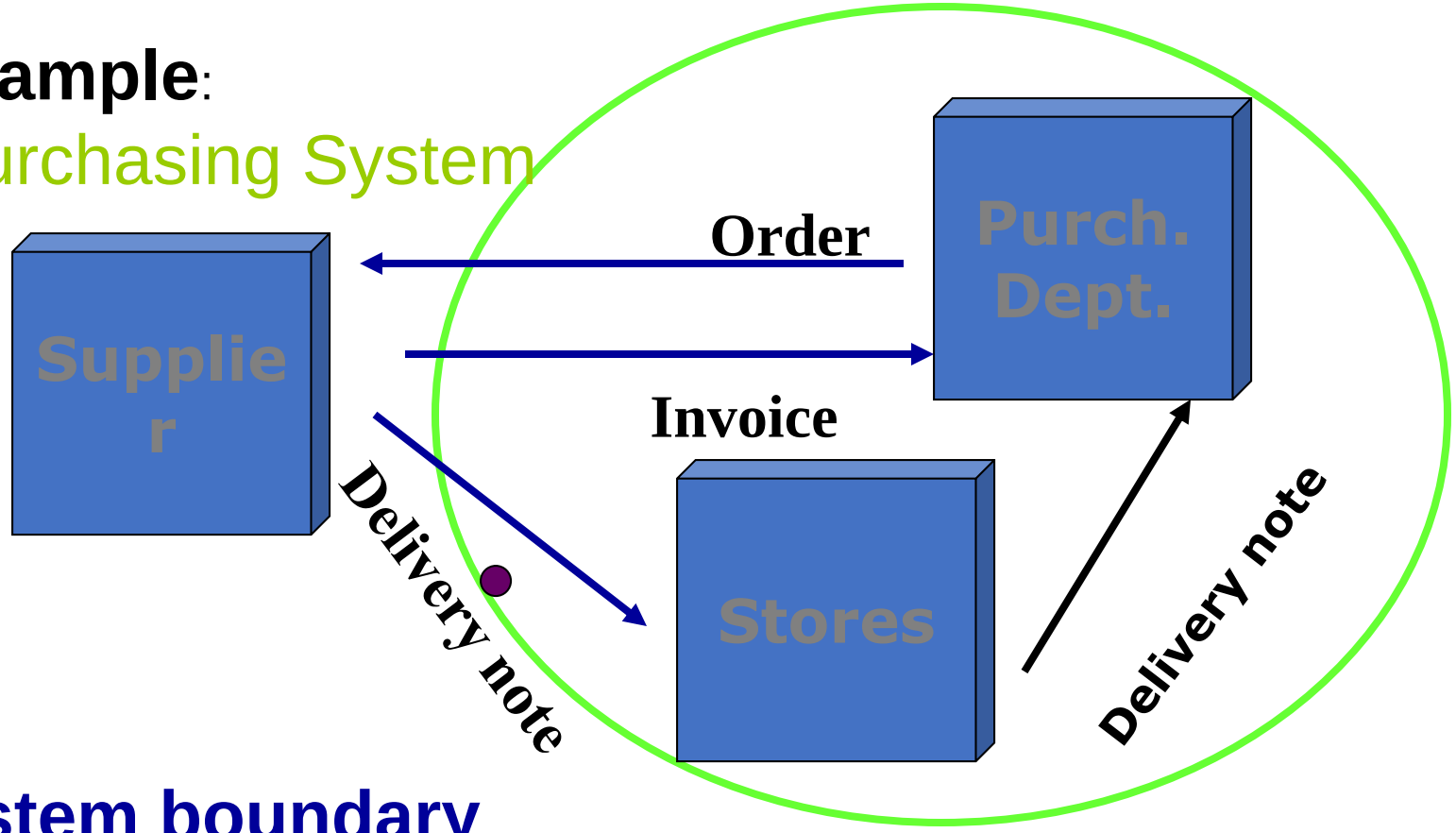


Example:
Purchasing System

Document flow diagram

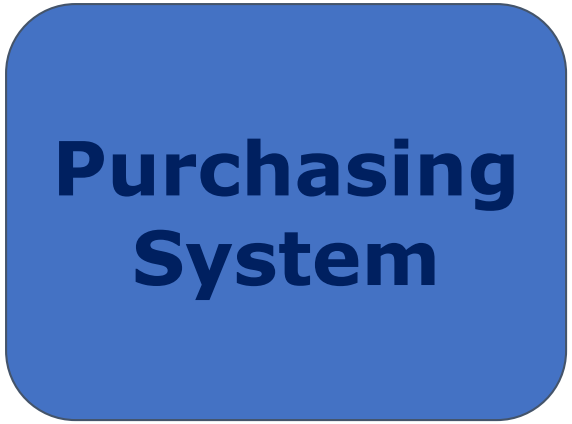
Example:

Purchasing System

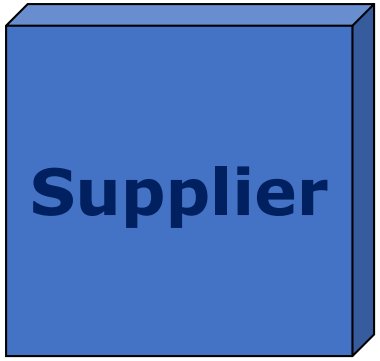


System boundary

Elements in a DFD
(Gane and Sarson Symbols)



A Process

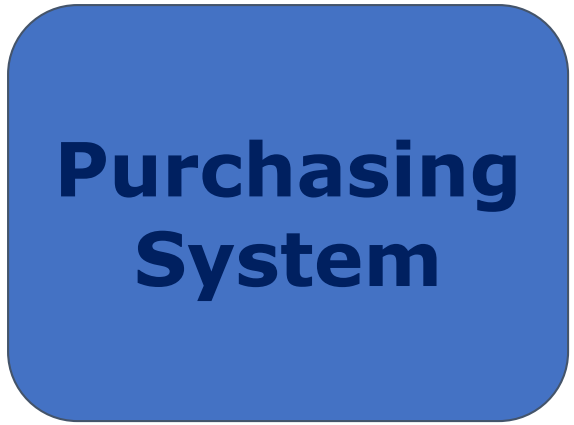


An External Entity

A Data Flow

A Data Store

Elements in a DFD

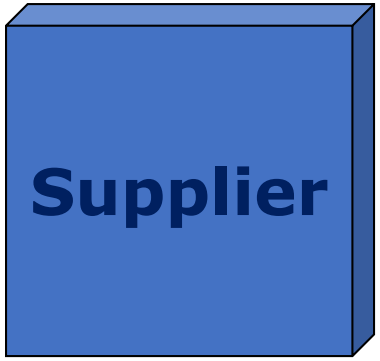


Function carries out by the business, which transforms or uses data

A Process or Business Function

Gane and Sarson Symbol

Elements in a DFD



Data Flow's origin or destination which is outside the business area being modeled.
Some times called 'Source' or 'Sinks'

An External Entity

Gane and Sarson Symbol

Elements in a DFD

A Data Store

Data at rest. Files, complex data structures or sets of data

Gane and Sarson Symbol

Invoice

A Data Flow

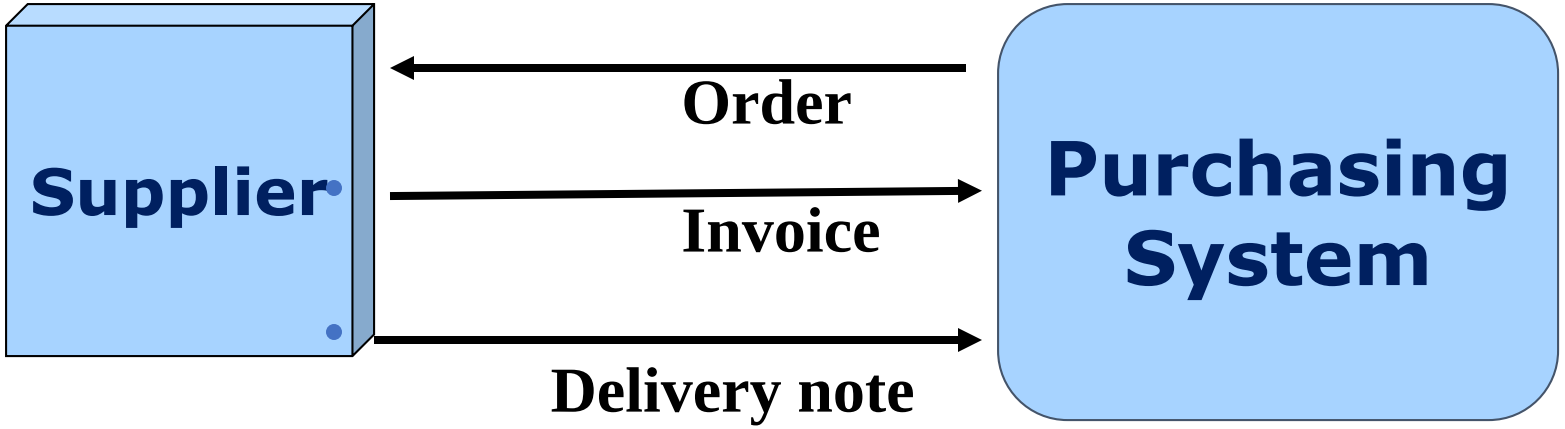
Indicates direction of data flow.

Represent single data elements or simple structures like records.

Gane and Sarson Symbol

Context Diagram

- Shows the top level function defined to cover the scope of the application.



How to Draw a Context Diagram

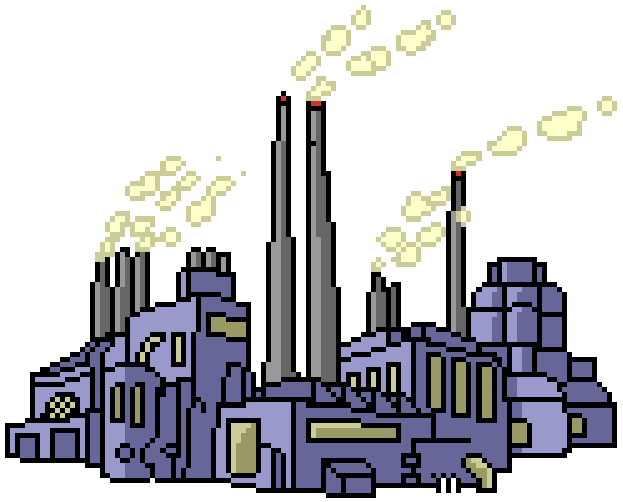
- Need to identify

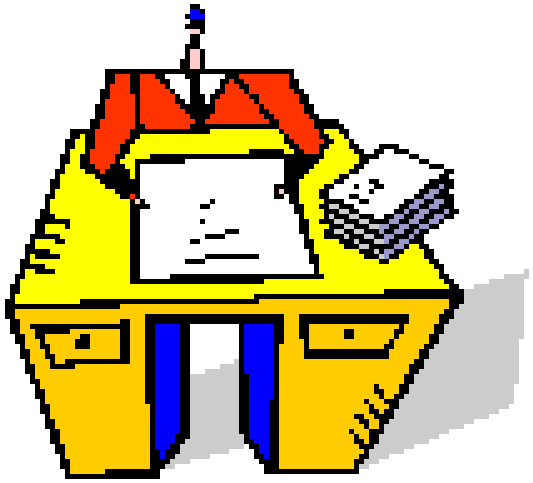
  the **data flows** and

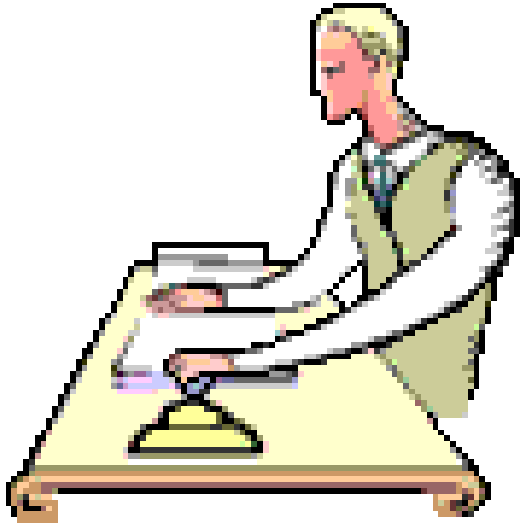
  the **external entities** needed for the context diagram

Task1

Identify all sources and recipient data to/from the system.

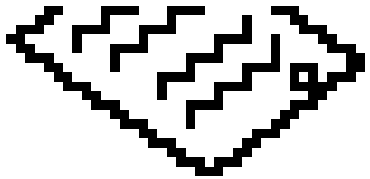






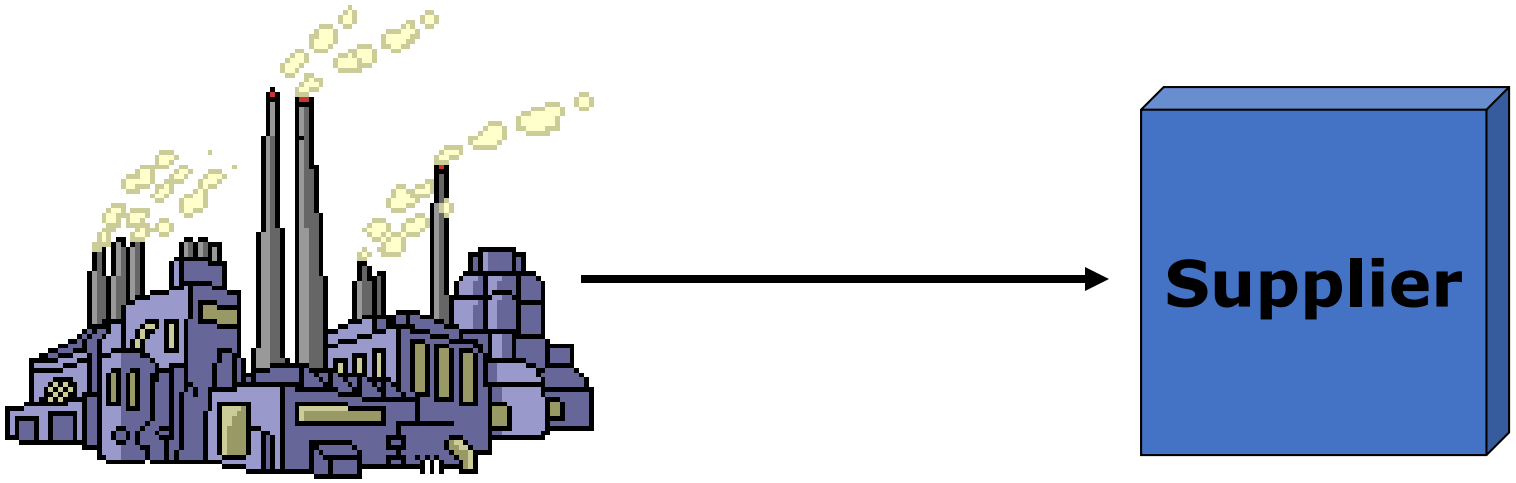
Task2

- Major data flows to and from the System



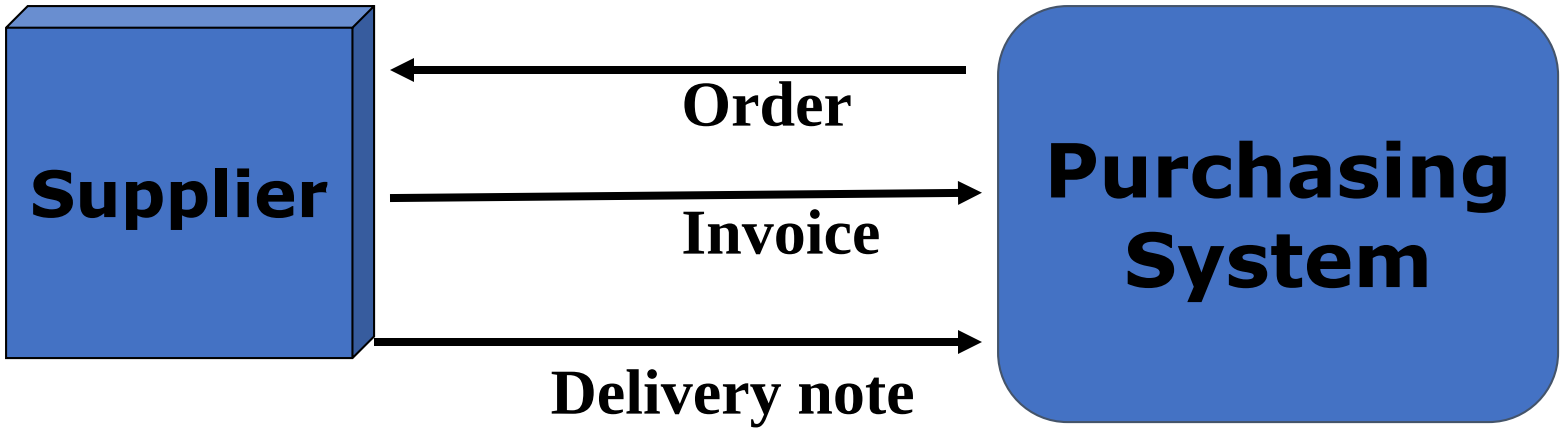
Task 3

- Convert each source or recipient into external entities



Task 4

- Add the data flows between each external entity and the process representing the entire system.



Data Flow Diagrams

- Draw Context Diagram
- Level 0 (Top Level) Data Flow Diagram
- Level 1 Data Flow Diagram
- Continue up to elementary functions

Bank Payment System

Consider a system in a bank whereby account holders get their withdrawals affected.

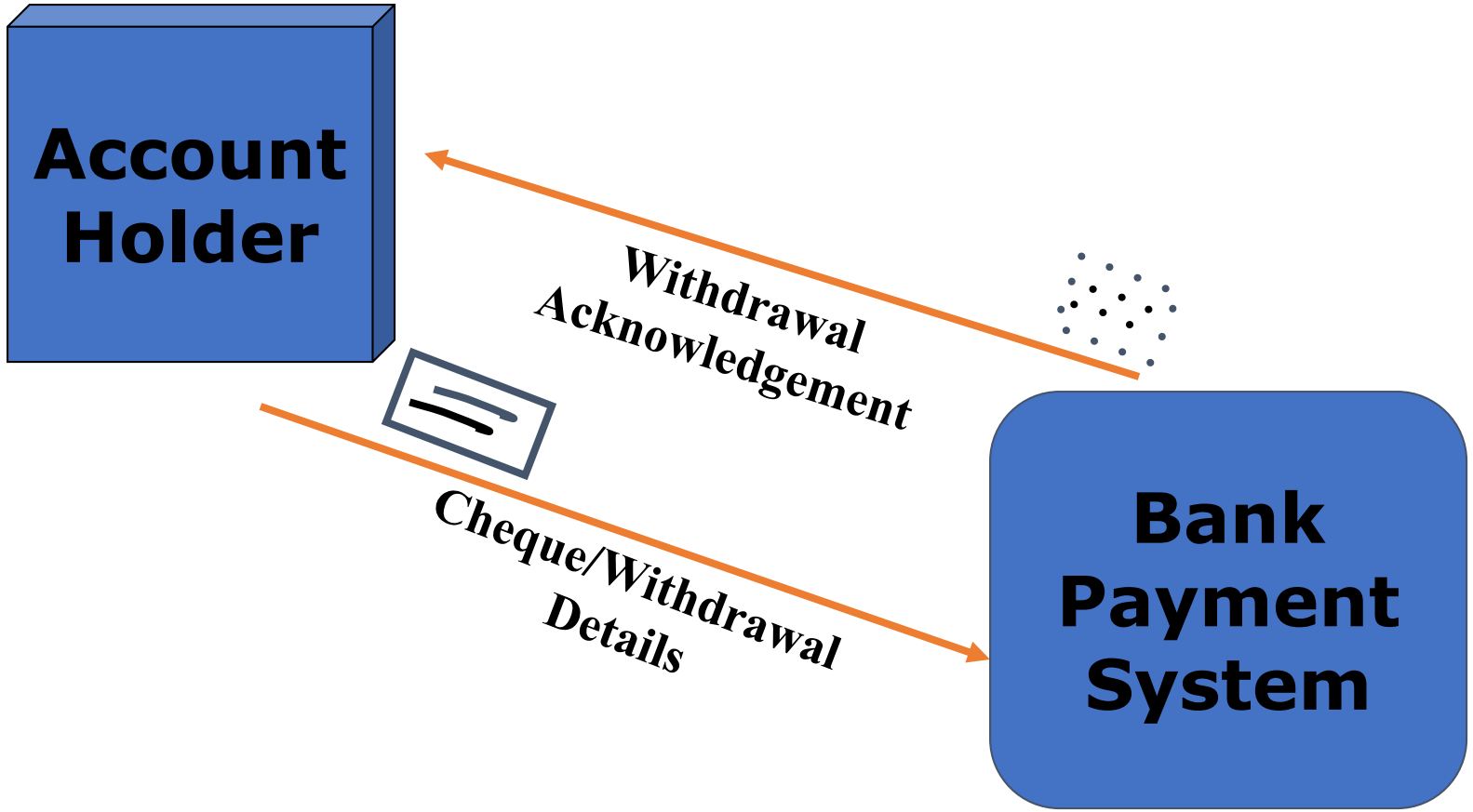
Whenever an account holder wants to withdraw some cash,

he presents a cheque or withdrawal slip.

Bank Payment System

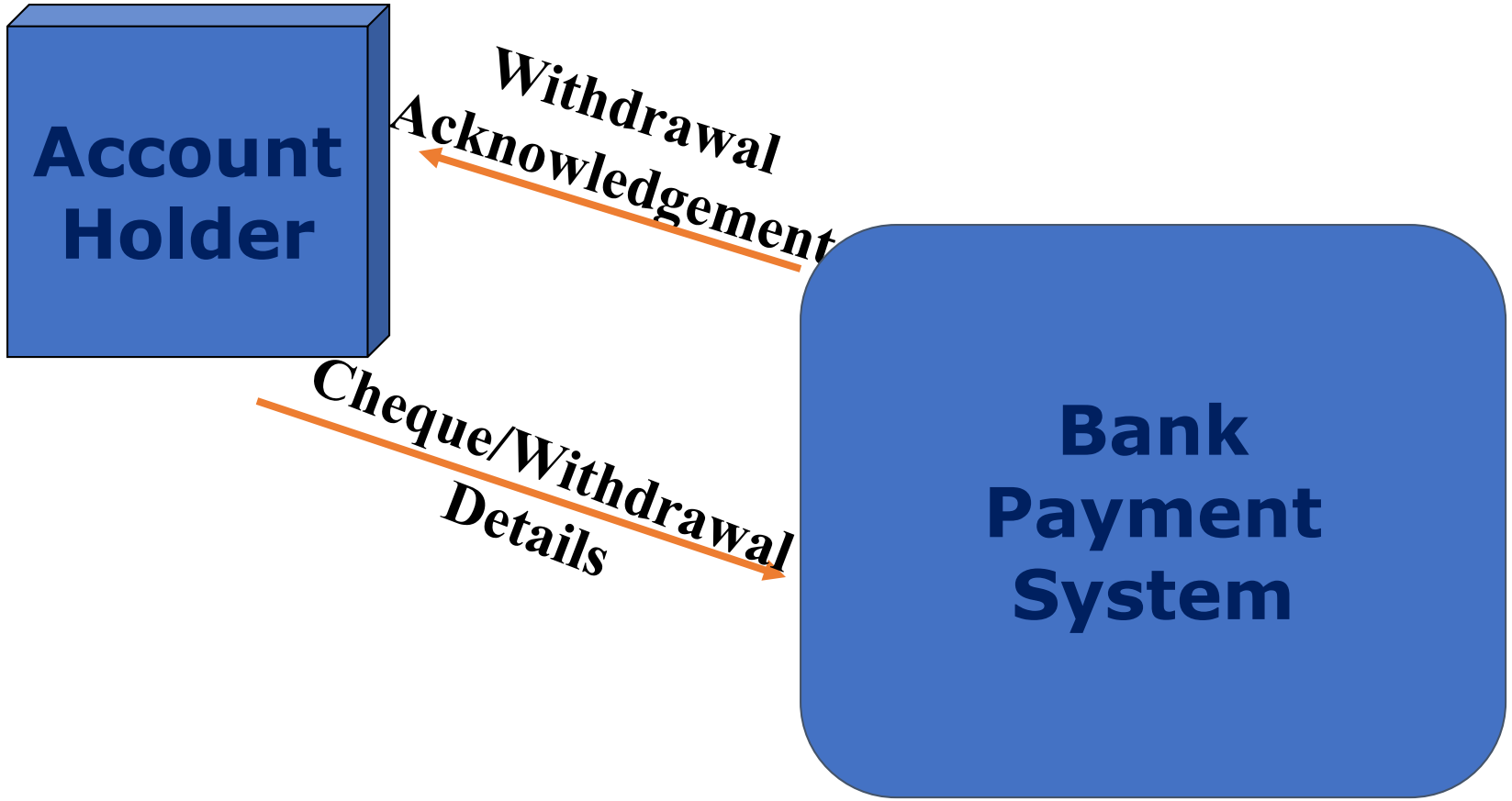
The account is checked for appropriate balance.

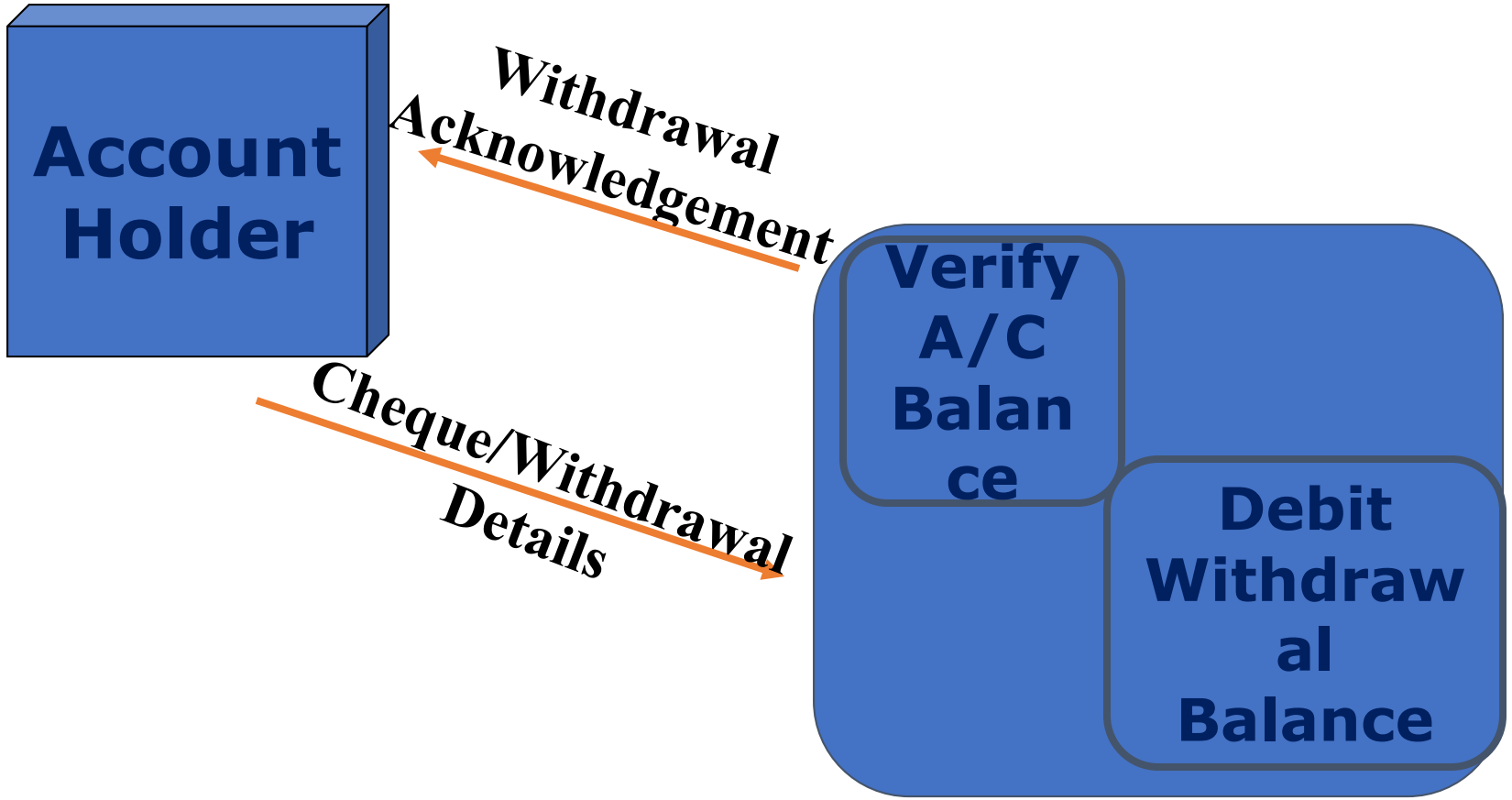
If balance exists, the cash is paid and the account is updated.

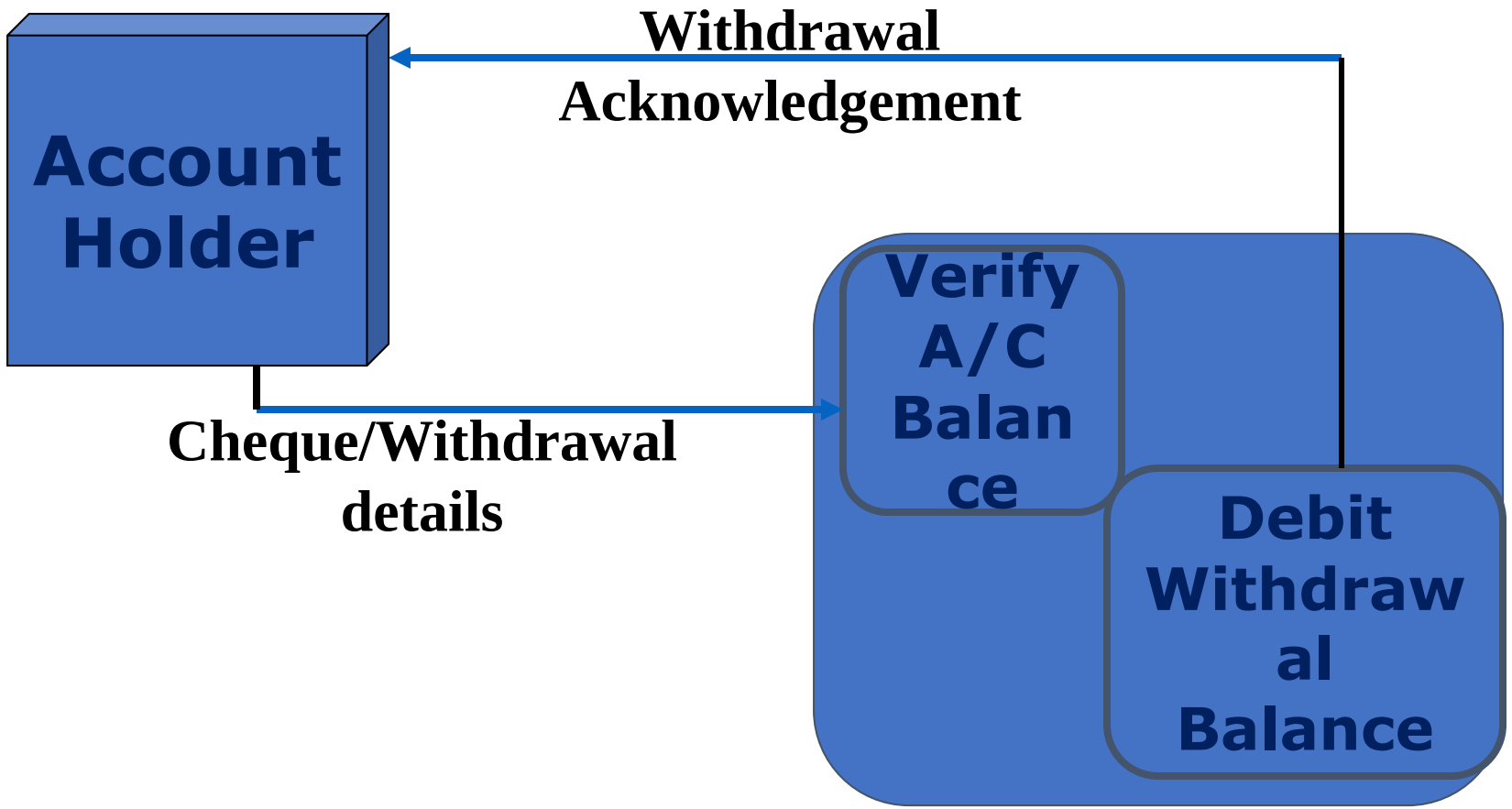

Account Holder
Withdrawal Acknowledgement
Cheque/Withdrawal Details
Bank Payment System

Decomposition

- is the act of breaking a system into its component subsystems , processes and sub processes.
- Top level function is then decomposed to its component functions
- Decomposition continues until the functions reach ‘elementary’ level.


Account Holder
Withdrawal Acknowledgement
Cheque/Withdrawal Details
Bank Payment System


Account Holder
Withdrawal Acknowledgement
Cheque/Withdrawal Details
Verify A/C Balan ce
Debit Withdraw al Balance


Withdrawal
Acknowledgement
Account
Holder
Verify
A/C
Balan
ce
Debit
Withdraw
al
Balance
Cheque/Withdrawal
details

Top Level DFD – Step3

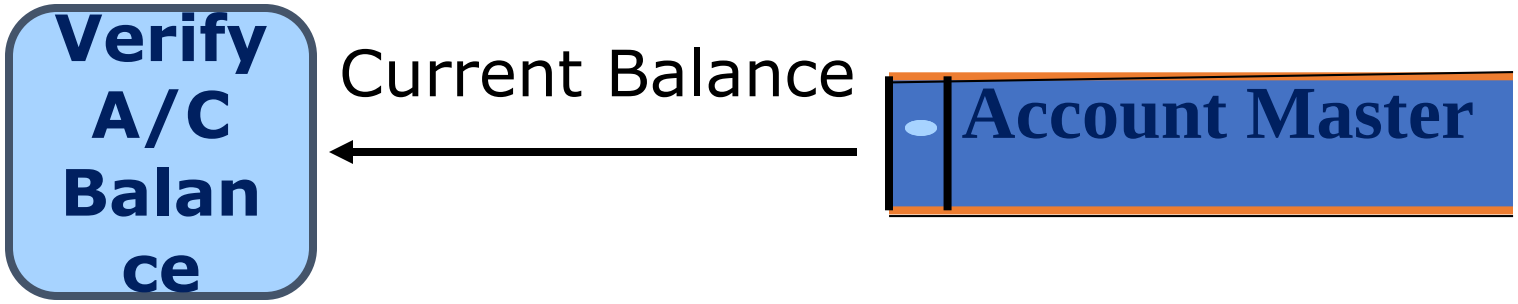
- Identify the Data Stores

Account Master

Top Level DFD – Step4

- Identify the other data flows.

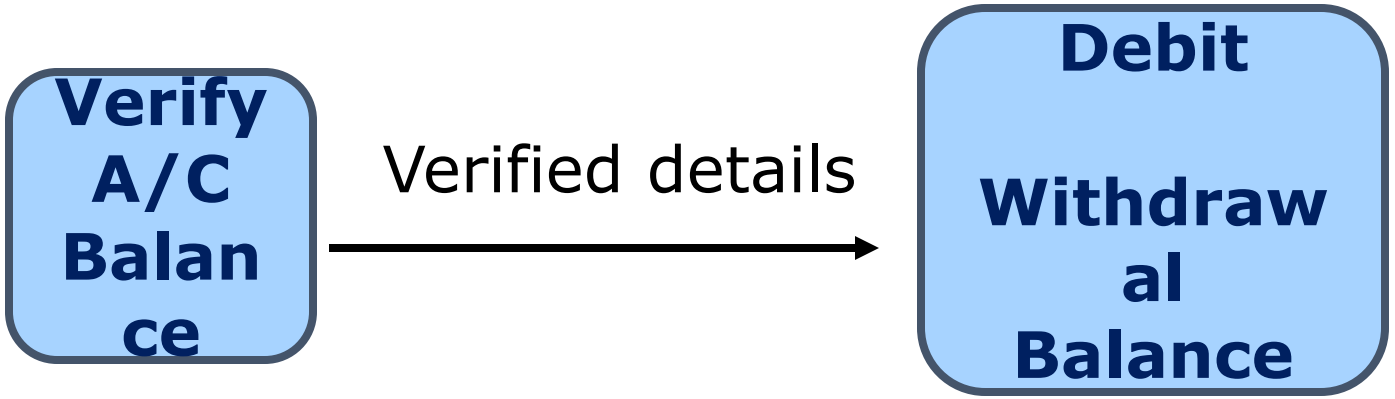
  Get current balance



Top Level DFD – Step4

- Identify the other data flows.

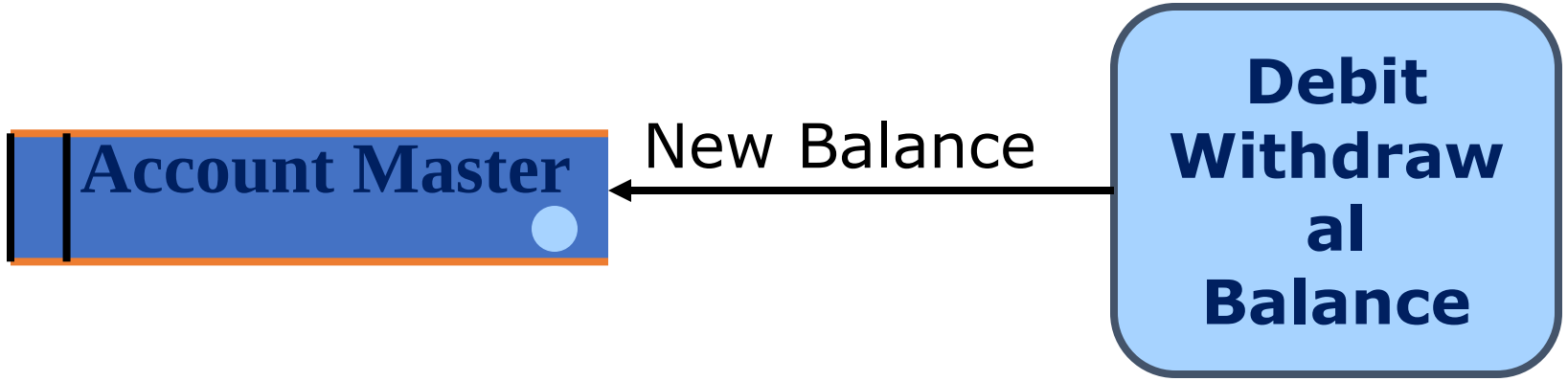
Transfer the verified details



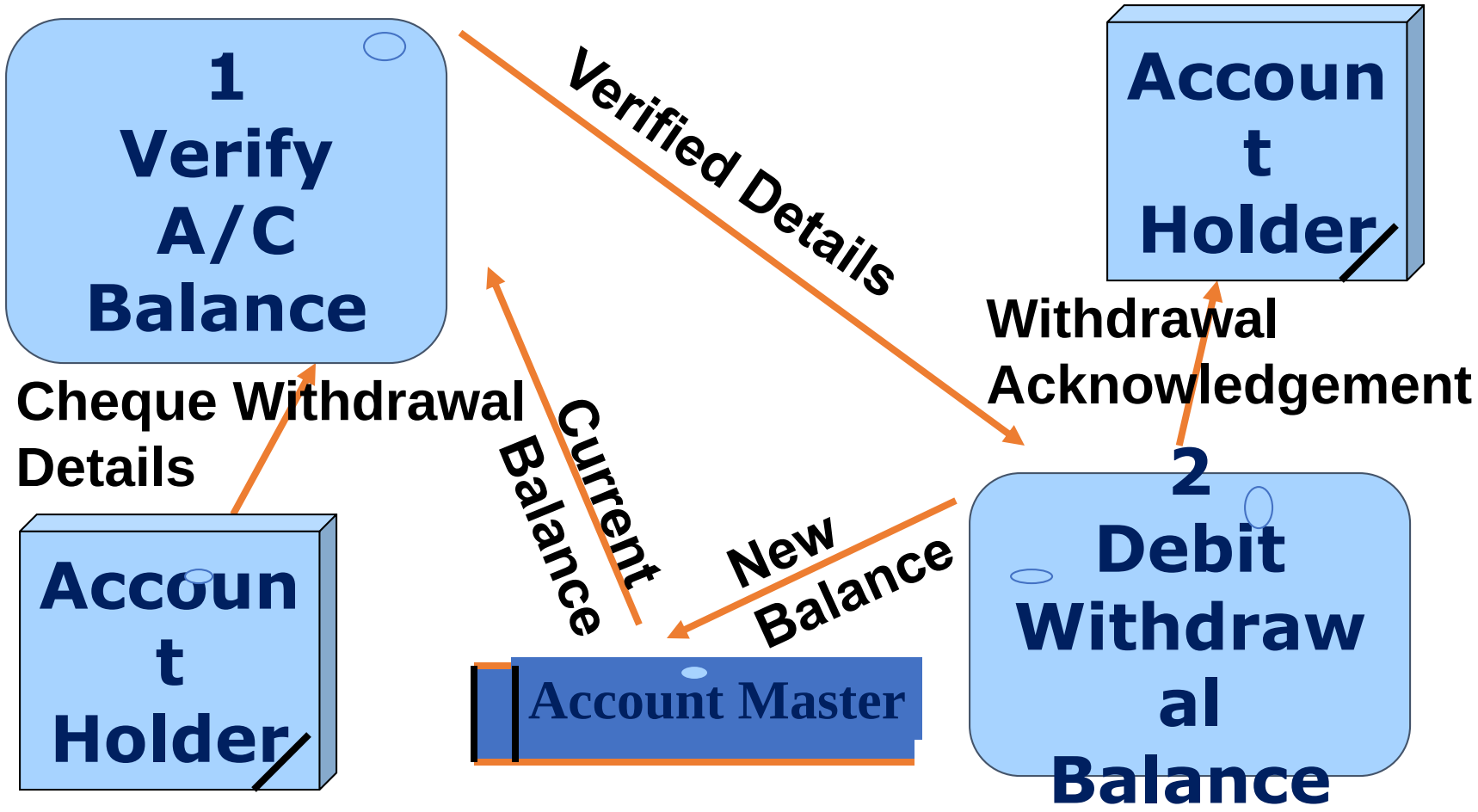
Top Level DFD – Step4

- Identify the other data flows.

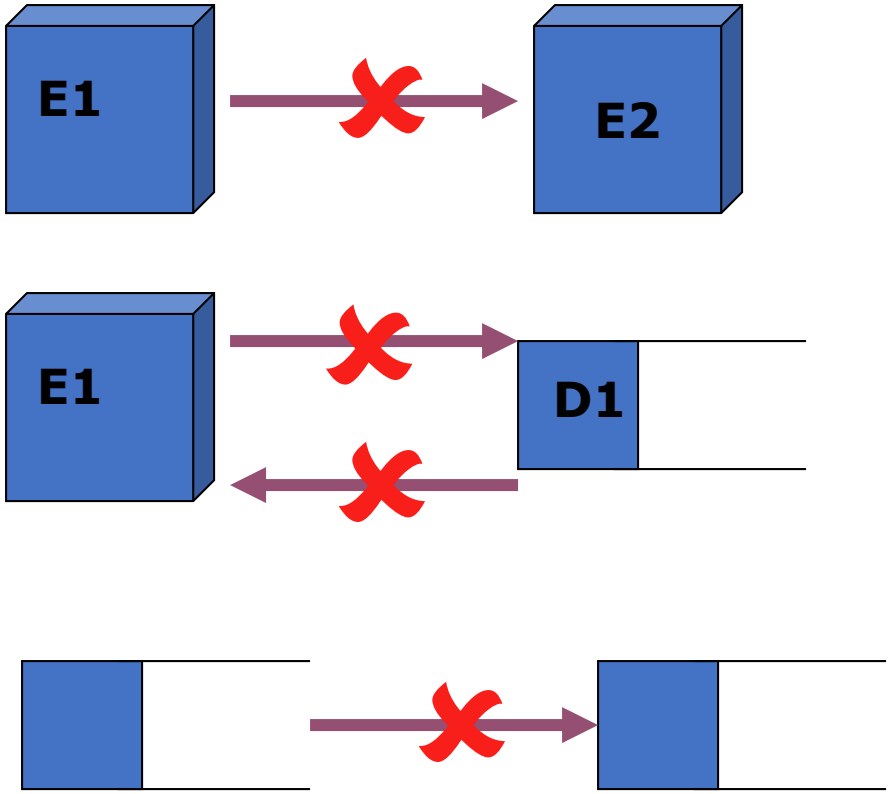
update new balance



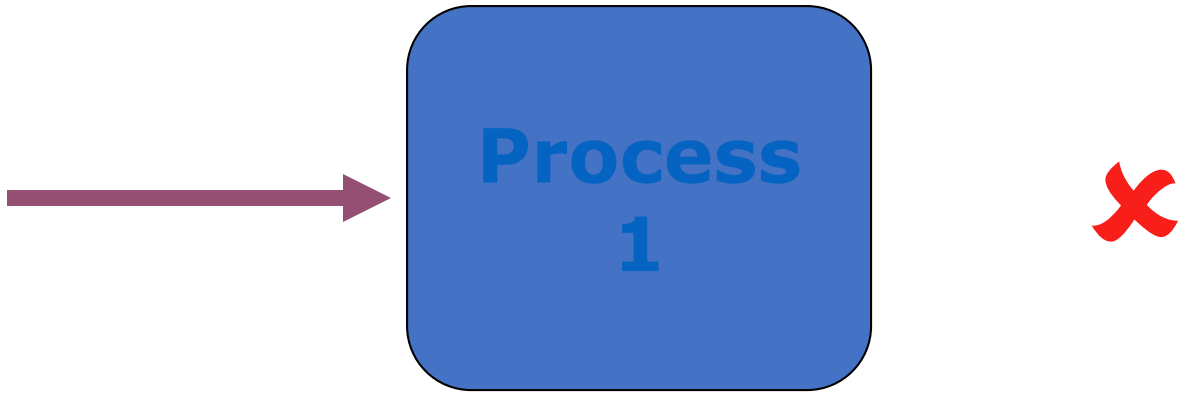
Top Level Diagram



Illegal Data Flows



Another Common error



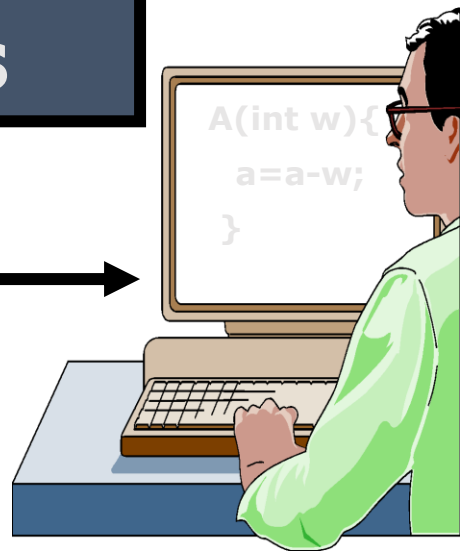
Read Only ?

Documenting Elements in DFD

- Element name is not enough.
- More important for elementary processes

You need to convert these into programs

Debit Withdrawal Balance



Methods of describing a Process

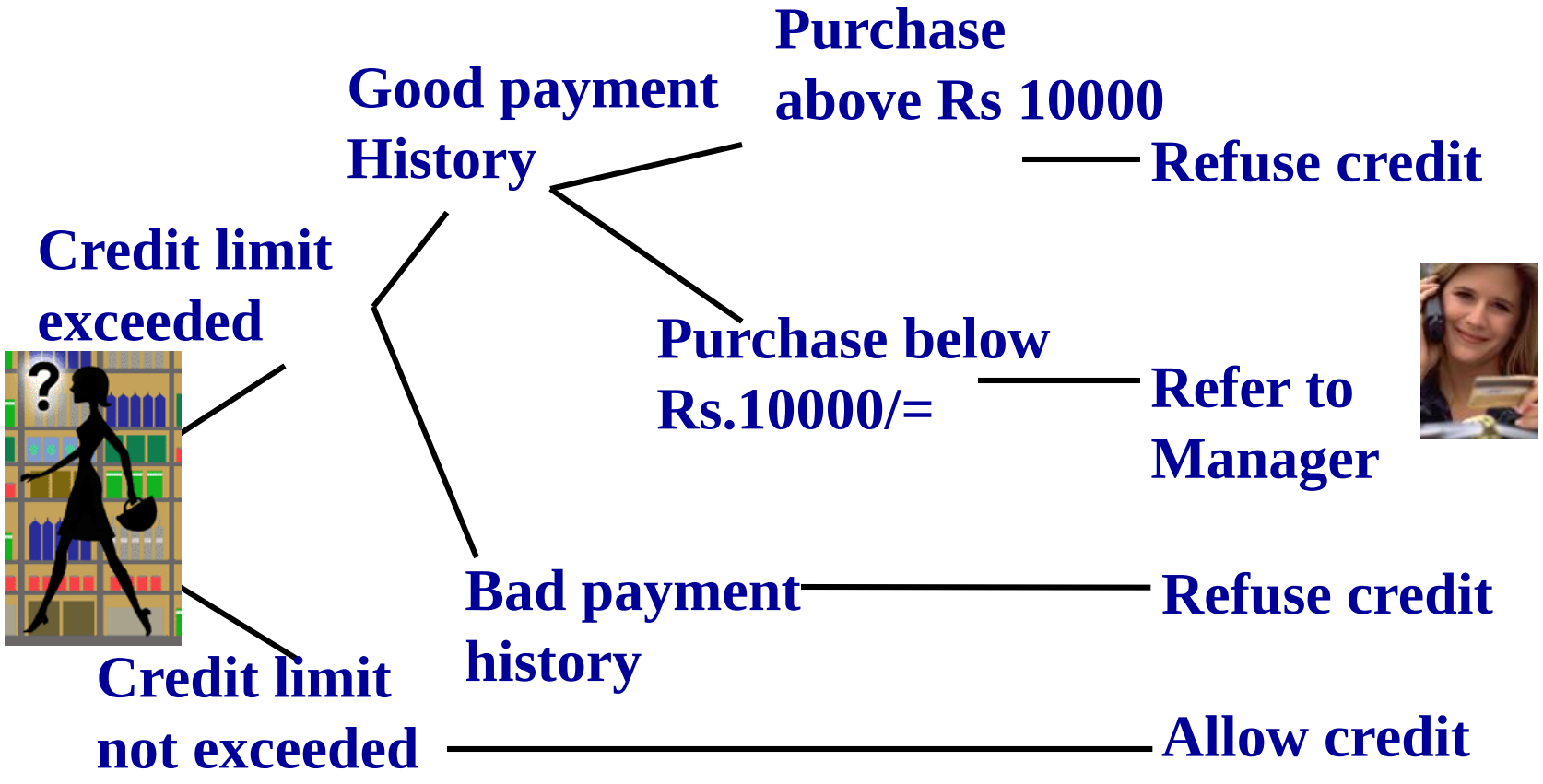
- **Structured English**
- **Decision Tree**
- **Decision Table**

Eg. A Process that has to determine whether a customer is to be given credit

Structured English

```
IF credit limit exceeded
THEN
      IF Customer has bad payment history
      THEN refuse credit
       ELSE
         IF purchase above Rs.1000/=
         THEN refuse credit
          ELSE refer to manager
ELSE  allow credit
```

Decision Tree



Decision Table

Condition	Credit limit exceeded	Y	Y	Y	Y	N	N	N	N
	Good payment history	Y	Y	N	N	Y	Y	N	N
	Purchase above Rs.1000/=	Y	N	Y	N	Y	N	Y	N
Action	Allow Credit								
	Refuse	X							
	Refer Manager		X						

Y-TRUE N-NOT TRUE X-TAKE ACTION

Decision Table

Condition	Credit limit exceeded	Y	Y	Y	Y	N	N	N	N
	Good payment history	Y	Y	N	N	Y	Y	N	N
	Purchase above Rs.1000/=	Y	N	Y	N	Y	N	Y	N
Action	Allow Credit					X	X	X	X
	Refuse	X		X	X				
	Refer Manager		X						

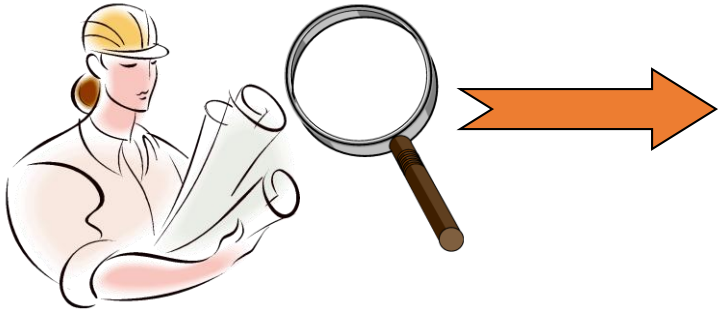
Y-TRUE N-NOT TRUE X-TAKE ACTION

System Design

- Decides ***how*** the system will operate,
  - in terms of the hardware, software, and network infrastructure;
  - the user interface, forms, and reports that will be used;
  - and the specific programs, databases, and files that will be needed.

Systems Design

- Produces a design specification for the new system. (Detail computer based solution)



Analysis

- Select equipment
- Design databases
- User Interface Design
- Design programs etc

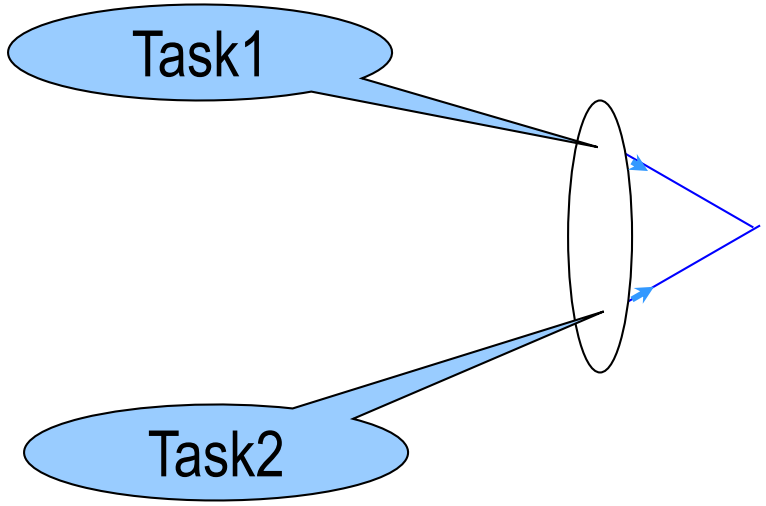
Design

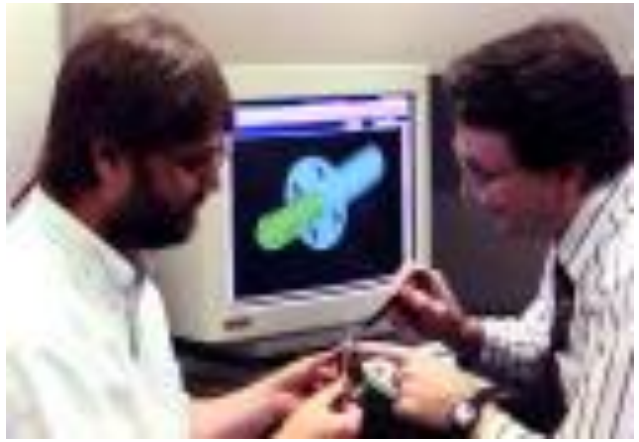
Systems Design

Systems analysis - emphasize on the business problem, *(What the system must do)*

Systems design - **focuses on the technical or implementation concerns of the system.**

(How to do?)





System Design Text book Definition

- Process of Defining
  - the components,
  - Interfaces,
  - database tables etc.

for a system to satisfy specified **functional** and **Non-functional** requirements

Library System

Typical Functional Requirements

- Lending/Issuing of Books
- Reservation of Books
- Returning of Books
- Member Registration
- Book/Copy Registration

Library System

Typical Non-Functional Requirements

- Performance Requirements
  - Need Quick Response Time
  - Should be able handle more than 10000 records
- Security
  - User Authentication and validation of members using a unique ID
- Safety : Must have two servers (one for backup)

System Design

Analysis Models

- Acts as a link between **system description** and **design model**.
- In Analysis Modelling, information, behavior and functions of the system is defined and translated into the architecture, component and interface level design in the design modeling.

Design Models

- The design models are used as essential input to activities in implementation and test.

Four main types

- Architecture Design
- Software Design
- Data base Design
- Interface Design